

impressive quantum processors using superconducting technology, achieving milestones like quantum supremacy and demonstrating the practical potential of quantum computing.

Today, the landscape of quantum hardware is diverse and dynamic. While superconductors lead the way, research is ongoing into improving their performance and scalability. At the same time, silicon is enjoying a resurgence, with new approaches seeking to combine the familiarity of silicon manufacturing with quantum capabilities—so-called ‘silicon qubits’. Hybrid platforms, merging different materials and technologies, are also being developed to address the specific needs of various quantum applications.

The evolution from silicon to superconductors reflects the ingenuity of scientists and engineers in overcoming technical hurdles and pushing the boundaries of what is possible. As new materials and techniques continue to emerge, the future of quantum hardware looks both exciting and unpredictable, with the promise of ever more powerful machines on the horizon.

The progress made in the field of quantum computing

There is a lot happening globally to make quantum computing mainstream as soon as possible. Here’s a quick look at the progress made so far.

Matured silicon spin qubit fabrication: Foundries are now producing silicon spin qubits with unprecedented uniformity and yield. This is achieved by adapting existing CMOS manufacturing processes, allowing for the creation of multi-qubit processors with consistent performance characteristics across the chip, a critical step for scalable quantum computation.

Enhanced qubit coherence in silicon: Researchers have successfully engineered the silicon lattice to be isotopically pure (Silicon-28), drastically reducing magnetic noise. This has pushed the coherence times

of spin qubits into the multi-second range, allowing for a significantly higher number of quantum operations to be performed before the quantum state decoheres.

High-fidelity two-qubit gates:

The precision of two-qubit gates, the fundamental building blocks of quantum algorithms, has surpassed the 99.9% fidelity threshold in silicon. This level of accuracy is the minimum required for effective quantum error correction, marking a transition from noisy intermediate-scale quantum (NISQ) devices to fault-tolerance.

Cryo-CMOS for integrated control:

Cryogenic CMOS (Cryo-CMOS) circuits are now being integrated directly on-chip or in the same package as the qubits. These specialised chips can operate at millikelvin temperatures, drastically reducing the number of wires needed to control the qubits and minimising thermal noise, which is a major source of errors.

Advanced 3D stacking and heterogeneous integration: We are seeing the first demonstrations of 3D-integrated quantum processors. This involves stacking layers of qubit arrays with layers of classical control and readout electronics, connected using through-silicon vias (TSVs). This approach allows for a massive increase in qubit density and functional integration.

On-chip quantum error correction (QEC) codes: Small-scale, hardware-efficient QEC codes are being implemented directly on semiconductor-based quantum chips. These codes can detect and correct certain types of quantum errors in real-time, extending the computational lifetime of the qubits and improving the reliability of quantum calculations.

Development of cryogenic memory:

To support the Cryo-CMOS controllers, specialised cryogenic memory (Cryo-RAM) is being developed and integrated. This allows control signals and measurement results to be stored

locally at low temperatures, reducing the latency associated with communicating with room-temperature electronics.

Photonic interconnects for chip-to-chip communication: The challenge of scaling beyond a single chip is being addressed with on-chip photonic interconnects. These convert quantum information from electrons (in spin qubits) to photons, allowing quantum states to be transferred between different quantum chips with low loss, a key technology for building larger, modular quantum computers.

Automated qubit calibration and tuning: As the number of qubits grows, manual calibration becomes impossible. AI and machine learning algorithms are now routinely used to automate the complex process of tuning and calibrating thousands of qubits, optimising their performance and keeping the quantum computer stable over long periods.

Monolithic integration of qubits and control logic: Beyond 3D stacking, researchers are achieving monolithic integration, where qubits and their control transistors are fabricated in the same silicon layer. This represents the ultimate level of integration, promising the most compact and efficient design for a quantum microprocessor, mirroring the architecture of a classical CPU.

Standardised quantum design kits: Semiconductor foundries are beginning to offer early-access process design kits (PDKs) for quantum components. These kits provide a standardised set of rules and models for designing silicon-based quantum circuits, enabling a broader range of designers to enter the field of quantum hardware development.

Improved materials for superconducting-semiconducting hybrids: There is significant progress in creating hybrid devices that combine the best of superconducting and semiconducting qubits. New material interfaces between superconductors and semiconductors are being developed to enable fast, high-fidelity